

What we built for Project ARES

- Our project
  - **Our design:** Two working Unity simulators (both VR-ready on a Meta Quest): a realistic, flight-sized habitat and the bigger sci-fi Lunar Habitat Commander. They share the same AI engine.



Realistic game



Sci-fi game



- 18 months
  - **Our design:** The game clock runs whole days and nights. The tabletop test (Bonus A) runs 72 hours, and the threshold check runs 70 missions of 3–6 days each.
- 6 crew
  - **Our design:** We fly 4 because the NASA HUNCH page asks for 2–4. The design scales to 6 (2 more quarters and pods, +2 wearable sets, and the engine works with any crew size).
- #1 risk
  - **Our design:** Mental health comes first in every system: the AI senses how people are doing, triages, and adapts the habitat.
- 0 outside
  - **Our design:** VR nature walks, a greenhouse, Earth-view windows, video calls home and an observation lounge.

Your mission

Design the psychological support architecture for a six-person lunar habitat on an 18-month mission. Your system must use AI, biosensors, voice analysis, and facial recognition to detect and respond to crew psychological states — and adapt the living environment in real time.

Detect

Biosensors, voice prosody, facial recognition, and behavioural tracking feed continuous data to the AI

Analyse

AI cross-references signal streams, triages severity, and builds a longitudinal baseline for each crew member

Respond

The habitat adapts — lighting, audio, temperature, scent, schedules, and social nudges — automatically and proportionally

- Detect
  - **Our design:** 9 sensors: wristband (heart rate, HRV), skin patch (GSR, cortisol), chest beacon (location), voice and face in shared rooms only, CO2, and task data. You collect them yourself with the health check-in (press I).



Realistic game: health check-in

- Analyse
  - **Our design:** A weight table, each person's own 14-day baseline, how long the signal lasts, and three tiers. Confidence drops when sensors fail.

- Respond
  - **Our design:** Light, audio, temperature, scent, air, schedule and social suggestions, in order of how serious it is. Every change can be undone.

Six conditions to address

Your system must detect and respond to all six.

Anxiety

Acute stress from confinement, pressure, or safety fears

Depression

Sustained low mood and withdrawal building over days

Interpersonal conflict

Crew tension, avoidance, and communication breakdown

Cognitive fatigue

Declining task performance from sustained high-stakes work

Autonomy deprivation

Loss of personal agency in a tightly scheduled environment

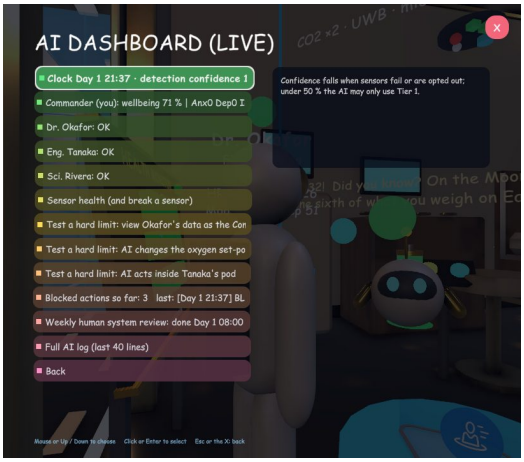
Sensory monotony

Perceptual habituation and boredom from unchanging surroundings

- All six conditions
  - **Our design:** All six are scored live for every crew member at the same time, so overlapping problems work too (board question Q2).



Realistic game: AI dashboard



Sci-fi game: AI dashboard

Sensor architecture

Phase 1 (Weeks 1–2): Research which sensors detect which conditions

HRV
Heart rate variability
Anxiety · stress · fatigue

GSR
Galvanic skin response
Acute anxiety · arousal

Sleep
Sleep architecture tracker
Depression · fatigue · circadian

Voice
Voice prosody analysis
Mood · conflict · fatigue

FACS
Facial action coding
Emotion · depression · conflict

Motion
Movement / proximity
Isolation · conflict · autonomy

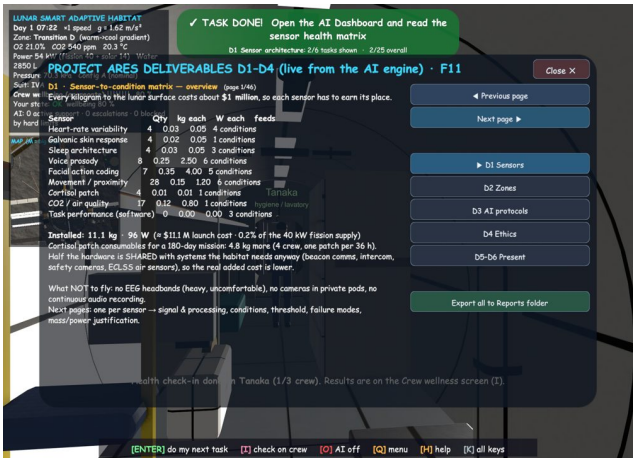
Cortisol
Cortisol wearable
Chronic stress · depression

Air
CO_2 / air quality
Cognitive fatigue · irritability

Key design question

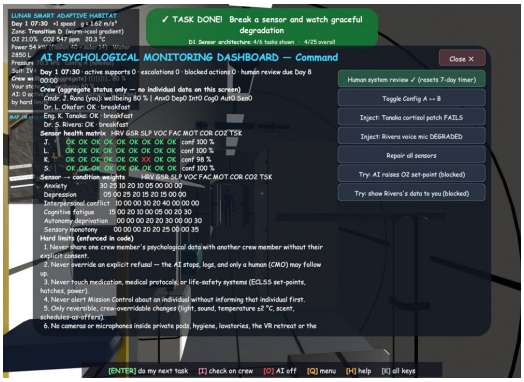
Every sensor added to a lunar habitat costs mass and power. Justify why each sensor is worth its weight. What happens when one fails? How does your system gracefully degrade?

- HRV, GSR, Sleep, Voice, FACS, Motion, Cortisol, Air
  - **Our design:** All 8 from this slide, plus task performance (software, 0 kg), make up our 9-sensor D1 matrix.

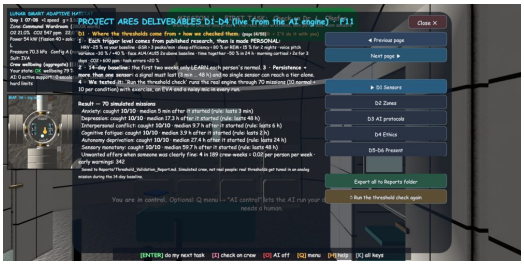


Realistic game: D1 matrix

- Key design question: mass, power, failure
  - **Our design:** About 13 kg and 110 W total. Each sensor has 2 failure modes, and when one breaks the AI leans on the others and stays at Tier 1 if it's less than 50 % sure.
    - We checked the thresholds with 70 simulated missions: all 60 real problems were caught and there were only 0.02 unwanted offers per person per week.



Realistic game: sensor failure



Realistic game: threshold check

Habitat spatial design

Phase 2 (Weeks 3–4): Design zones for psychological purpose, not just function

Communal zone	Private retreat	Work & mission bay
Shared meals, conversation, group activities. Warm lighting to draw crew together.	Acoustically shielded. Each crew member's personal space for rest and decompression.	Cool 6500K lighting, structured airflow. Optimised for sustained cognitive performance.
Exercise bay	**Nature / growth bay**	**Transition corridor**
Energising light and audio auto-activate. Screens show Earth views. Movement is medicine.	Plant growth area. Green light spectra, petrichor scent. Behavioural activation for depression.	Lighting and temperature shift between zones create psychological cues for mode-switching.

Deliverable: Annotated floor plans for two configurations — nominal operations and conflict protocol

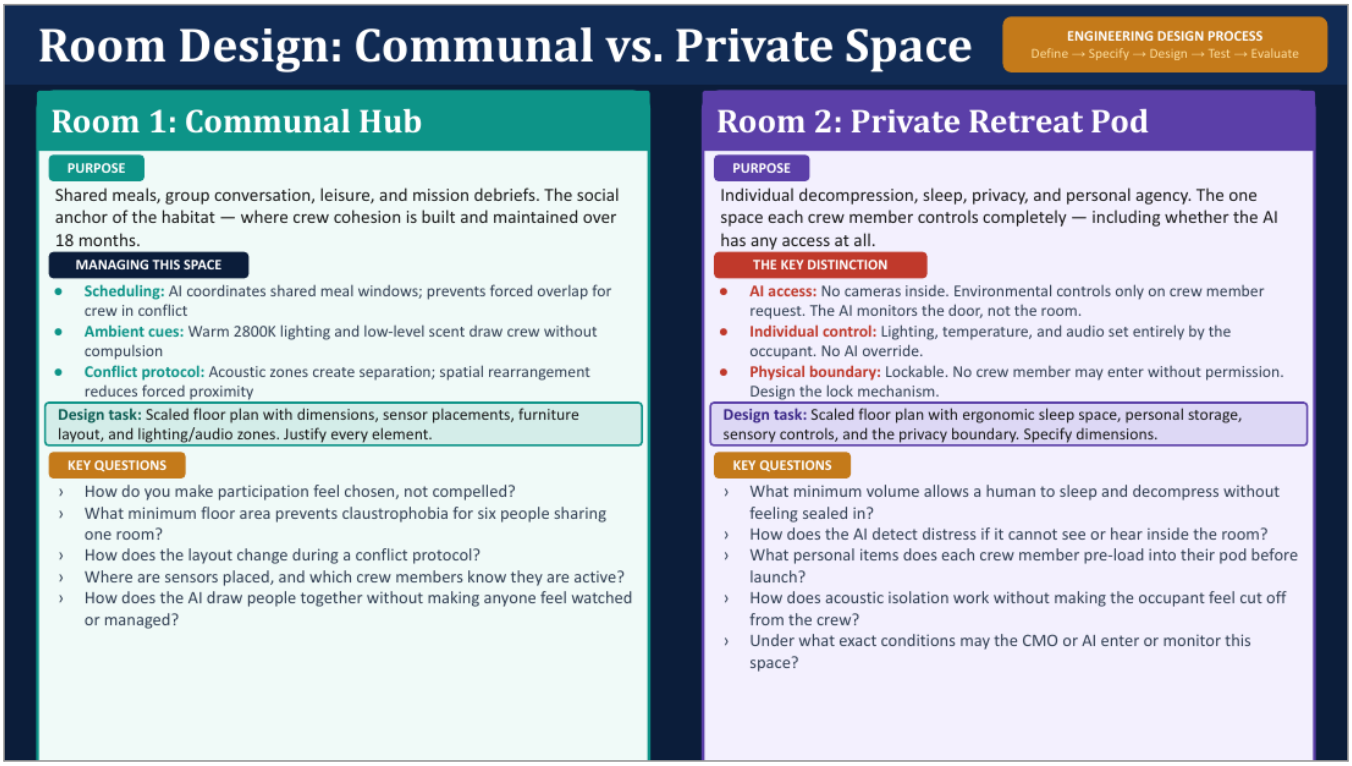
- Communal, Private retreat, Work & mission bay
  - **Our design:** Warm 2800 K galley and commons, private quarters with no cameras, and a cool 6500 K work zone with more airflow.
- Exercise bay, Nature / growth bay, Transition corridor
  - **Our design:** An exercise + VR room, a greenhouse with birdsong and green light, and hallways whose lights fade between zones (13 of them in the sci-fi base).
- Deliverable: annotated floor plans, two configurations
  - **Our design:** Dimensioned plans drawn to scale in both games, plus Config A and Config B (sound screens, second dining spot, second route).



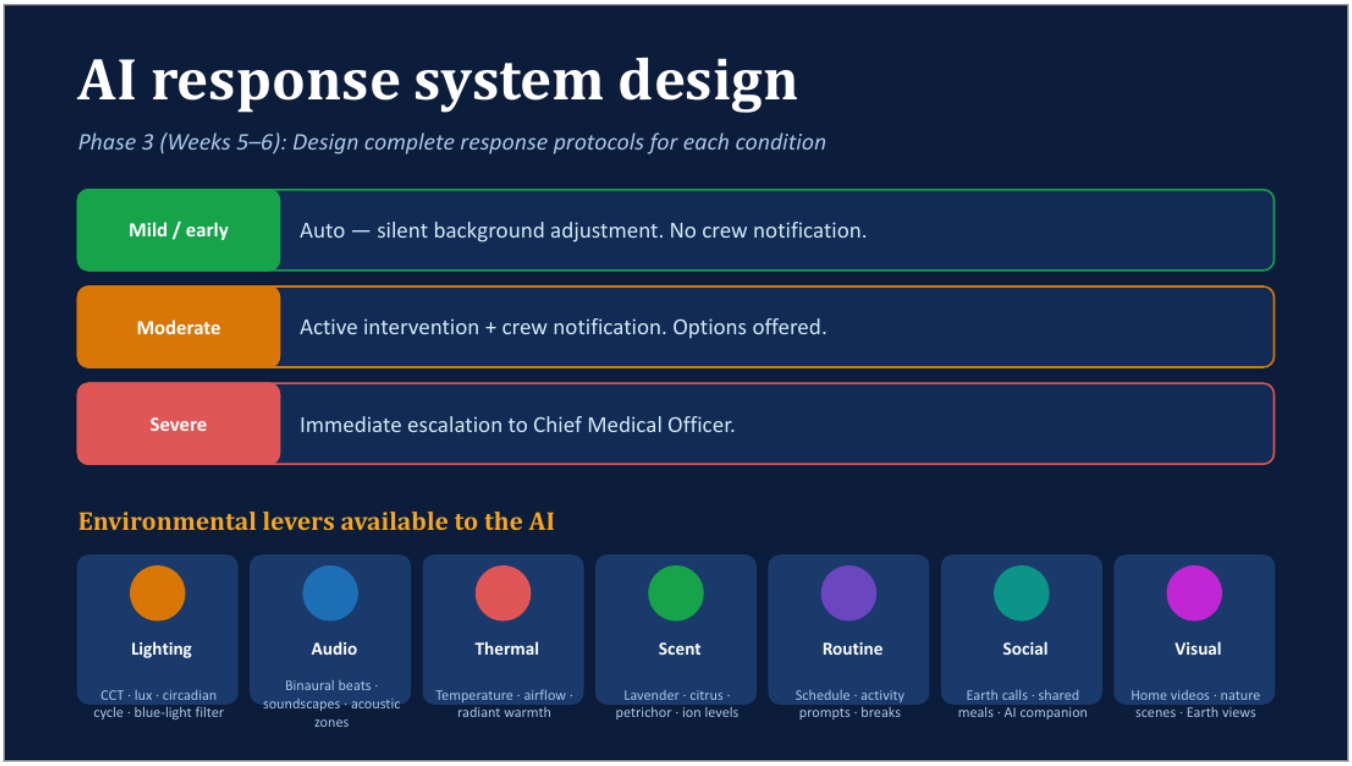
Realistic game: dimensioned plan



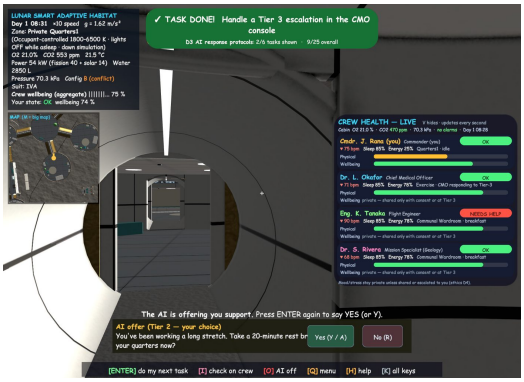
Sci-fi game: site plan



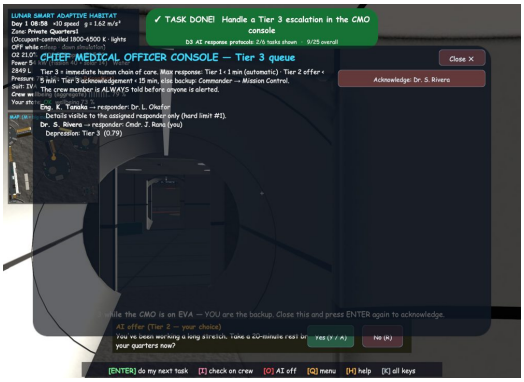
- Room 1: Communal hub
  - **Our design:** Core hub galley/wardroom (9.9 m upper deck). The AI times shared meals and never forces overlap for people in conflict. Warm light draws people in.
- Room 2: Private retreat pod
  - **Our design:** Crew quarters (8.8 × 8.8 m) plus a 2.1 × 2.1 m sleep pod. No cameras or mics, only a door sensor. The person controls light, temperature and sound, and the AI comes in only when invited.
- Emergency override
  - **Our design:** Only for life safety (alarms) or a Tier 3 case handled by the CMO. It's logged and the person is told.
- Key questions
  - **Our design:** Participation feels chosen because meals are offers, not orders. Volume: about 740 m³ per crew member overall and about 185 m³ of private space each. Distress without cameras is picked up by wearables and the door sensor.



- Mild / Moderate / Severe
  - **Our design:** Tier 1 silent (logged), Tier 2 offer you answer yes or no, Tier 3 tells the person first and then the CMO console.



Realistic game: Tier 2 offer



Realistic game: Tier 3 CMO console

- Environmental levers
  - **Our design:** All six domains are used, with a set order for each condition.

Example: Anxiety response flowchart

Students must produce flowcharts for at least 3 of the 6 conditions — here's the structure to follow

Signal detected → HRV drop · elevated HR · tense vocal tone · disrupted sleep

Severity triage → Score 4 signal streams · weight by duration · mild / moderate / severe

Auto intervention → Dim to 2400K amber · alpha binaural beats · raise temp to 23°C · lavender scent

Monitor 20 min → Track HRV · HR · vocal tone · movement patterns

Improving? → Yes → restore gradually · log event; No → escalate to CMO · crew check-in

Speed matters for anxiety

Anxiety spikes fast. The AI must act within minutes. Monitor window is 20 min — not 20 hours.

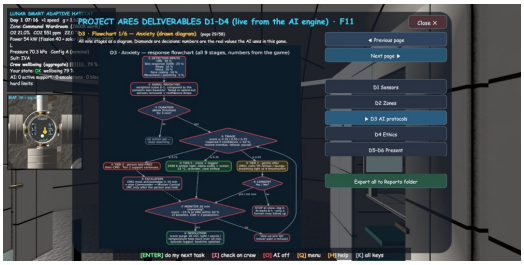
Silent first, then escalate

Most calming interventions work better when the crew member isn't aware they're happening.

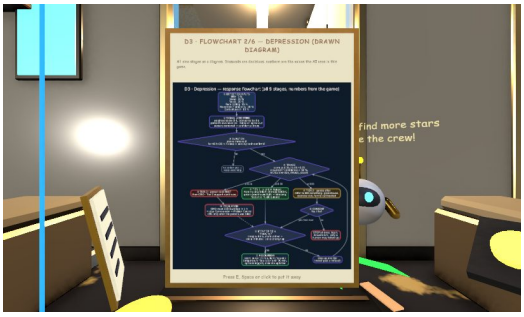
Never override consent

If a crew member explicitly refuses an intervention at any tier, the AI must stop and log.

- Example: anxiety flowchart
  - **Our design:** We drew flowcharts for all 6 conditions, with all 9 stages, inside both games. Anxiety matches this slide: dim to 2400 K, alpha audio, 23 °C, lavender, a 20-minute monitor, then restore or escalate.

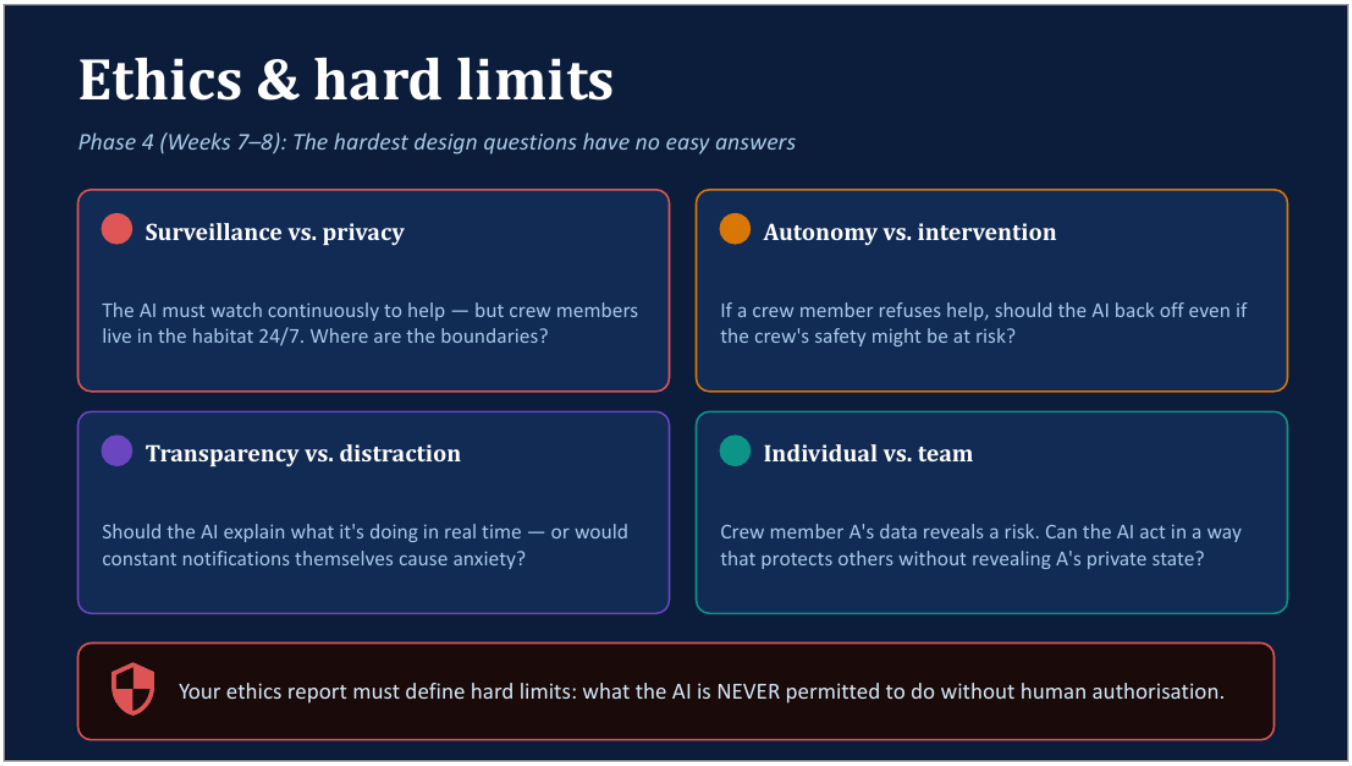


Realistic game: anxiety flowchart

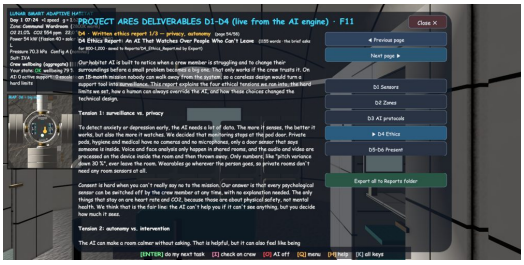


Sci-fi game: depression flowchart

- Never override consent
  - **Our design:** A "no" stops the AI at once, it's logged, and it waits 6 hours. Only a human can follow up.



- Four tensions + hard limits
  - **Our design:** Our D4 report (1,155 words) answers all four. The 8 hard limits are enforced in code, and every crew member can turn the AI off (O / F12).



Realistic game: D4 report



Realistic game: hard limits

- Night watch
  - **Our design:** If anything goes wrong while the crew sleeps, the AI stops itself, undoes its changes and wakes a human, who is then in charge.

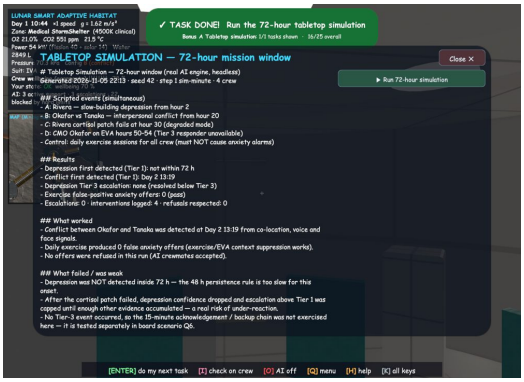


Realistic game: night watch



- Required deliverables
  - **Our design:** D1–D6 are all in both games (F11). The journals are written separately by each of us.

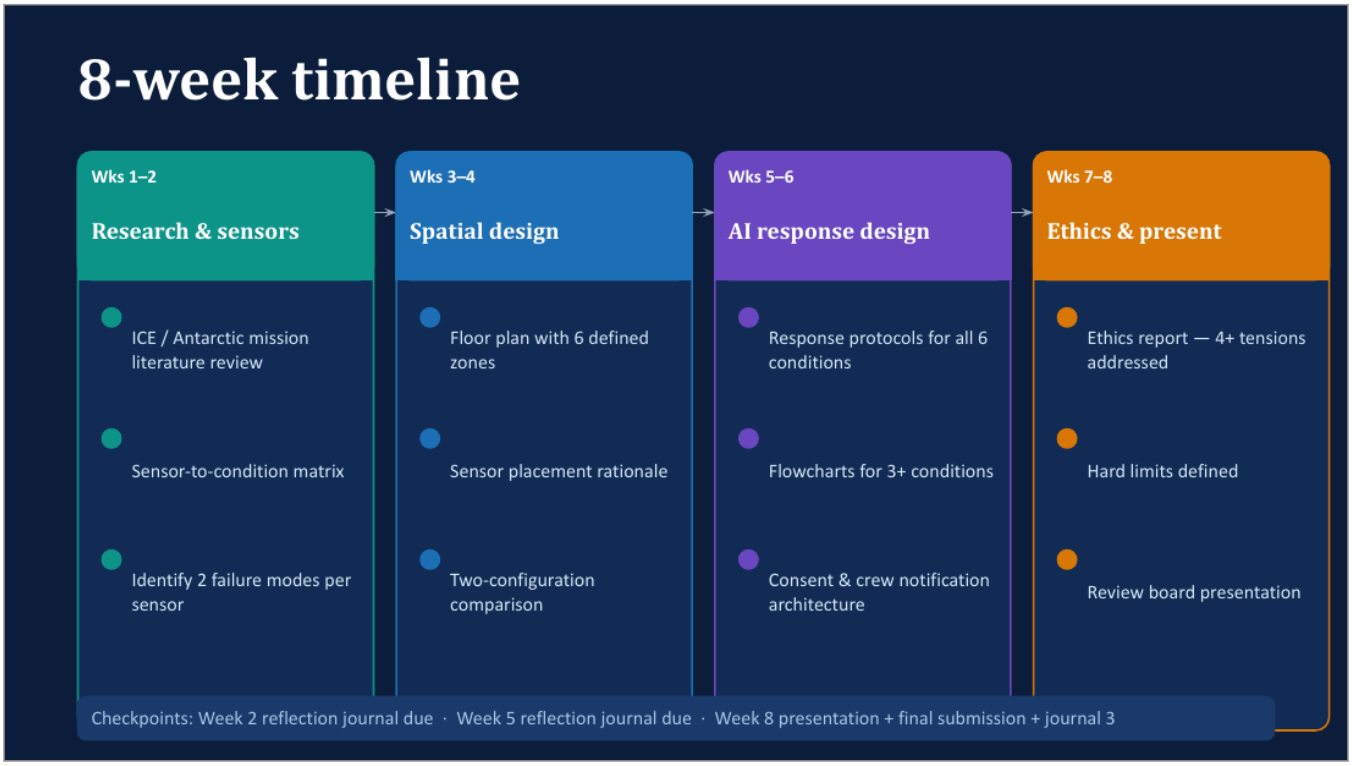
- Points breakdown (240 total)
  - **Our self-grade on this 240-point scale: 240 / 240**
    - AI response protocols 50/50 · Ethics 40/40 · Sensor architecture 35/35 · Spatial design 35/35
    - Presentation 25/25 · Journals 15/15 · Tabletop simulation 20/20 · Interface mockup 20/20



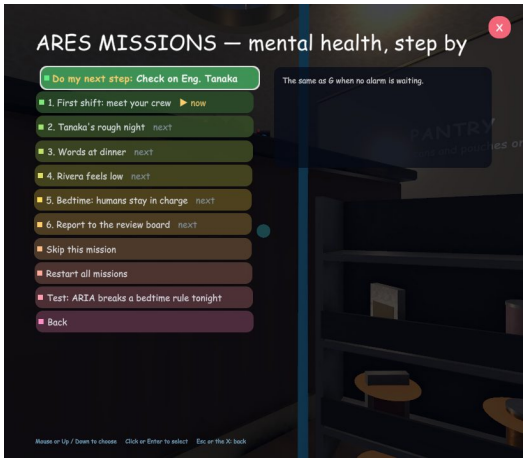
Realistic game: tabletop (Bonus A)



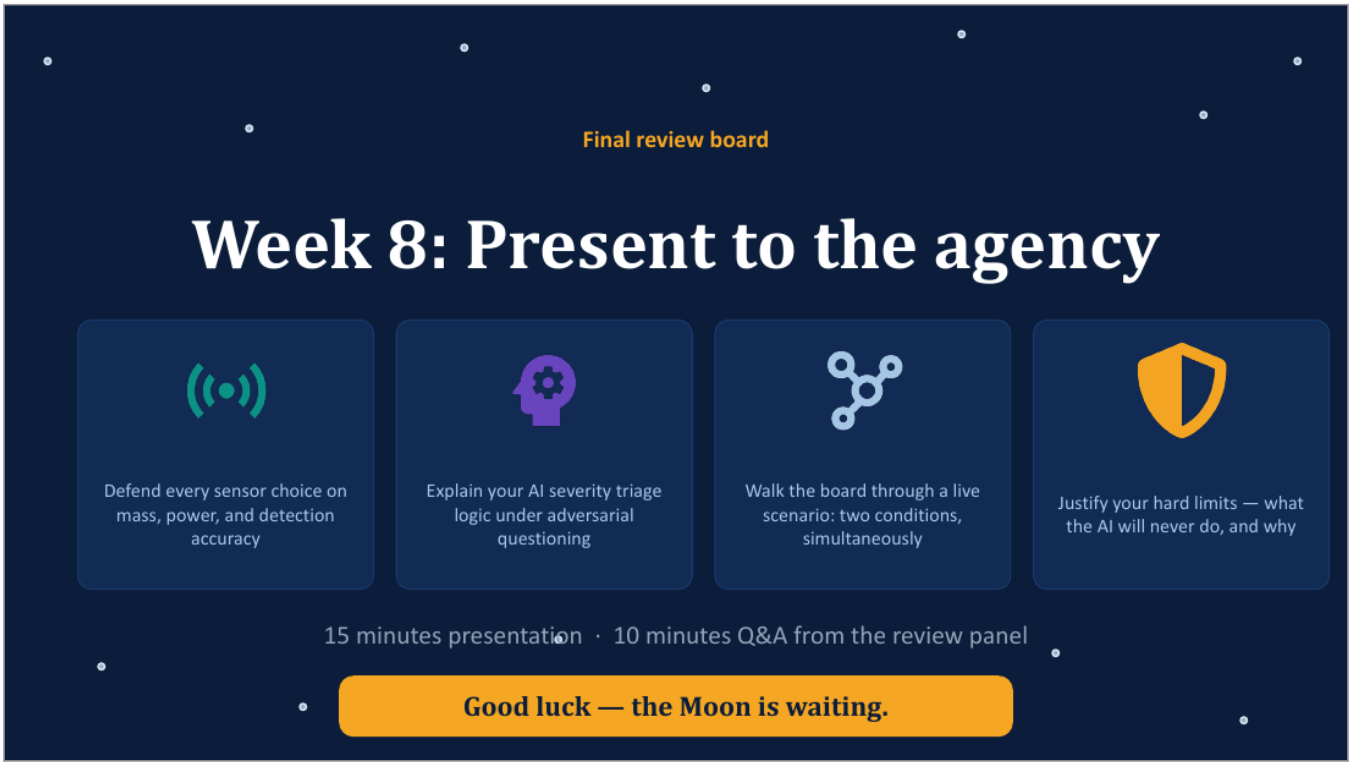
Realistic game: crew interface (Bonus B)



- 8-week timeline
  - **Our design:** We followed it. The game's missions follow the same order (sensors → spaces → protocols → ethics and the board).



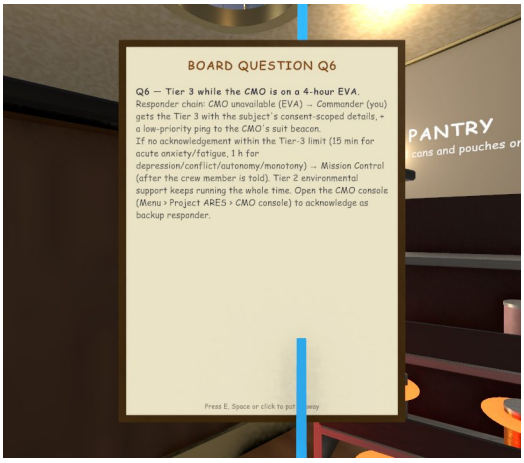
Sci-fi game: missions



- Final review board
  - **Our design:** Every board question (Q1–Q6) runs live in both games, including the two-conditions-at-once scenario (Q2) and the CMO-on-EVA backup (Q6).



Realistic game: Q2



Sci-fi game: Q6